Case Study 7: Campus Registration Identifier Analyzer

Aim: To develop a Python program that examines student identification strings, validates their structure, extracts important components, and classifies student records according to branch codes and admission year.

1. Problem Statement

A university assigns a unique registration ID to every student. The ID contains information such as admission year, branch code, and student number.

The program should:

1. Accept student registration IDs.

2. Validate the ID format.

3. Extract the admission year.

4. Extract the branch code.

5. Extract the student number.

6. Classify students according to their branches.

7. Display students according to admission year.

8. Reject invalid registration IDs.

9. Assumed Registration ID Format

Registration ID format:

YYYYBRNNN

Component	Description	Example
YYYY	Admission year	2026
BR	Branch code	CS
NNN	Student number	101

Example: 2026CS101

Branch codes:

Code	Branch
CS	Computer Science
EC	Electronics
ME	Mechanical
EE	Electrical
IT	Information Technology

3. Algorithm

4. Start.

5. Enter the number of student records.

6. Read each registration ID.

7. Check whether the ID contains exactly 9 characters.

8. Check whether the first four characters are digits.

9. Check whether the next two characters represent a valid branch.

10. Check whether the last three characters are digits.

11. Extract the admission year, branch code, and student number.

12. Classify students according to their branch.

13. Display valid and invalid records.

14. Stop.

15. Pseudocode

START

INPUT number_of_students

CREATE empty student list

```
FOR each student

    INPUT registration_id

    IF length of ID = 9 AND
       first 4 characters are digits AND
       last 3 characters are digits AND
       branch code is valid

        EXTRACT admission year
        EXTRACT branch code
        EXTRACT student number

        STORE student details

    ELSE

        DISPLAY Invalid registration ID

    END IF

END FOR

DISPLAY valid student records

CLASSIFY students by branch
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DISPLAY admission year and branch details

STOP

5. Python Program

Complete Python solution

```
# Campus Registration Identifier Analyzer# Branch codesbranches = {    "CS": "Computer Science",    "EC": "Electronics",    "ME": "Mechanical",    "EE": "Electrical", "IT": "Information Technology"}students = []invalid_count = 0n = int(input("Enter number of student records: "))# Process student registration IDsfor i in range(n): print("\nStudent", i + 1)    reg_id = input("Enter registration ID: ").upper()    # Validate registration ID    if (len(reg_id) == 9 and        reg_id[:4].isdigit() and        reg_id[4:6] in branches and        reg_id[6:9].isdigit()):        # Extract components        year = reg_id[:4] branch_code = reg_id[4:6]        student_number = reg_id[6:9]
```

6. Sample Output

Enter number of student records: 4

Student 1

Enter registration ID: 2026CS101

Valid Registration ID

Admission Year: 2026

Branch: Computer Science

Student Number: 101

Student 2

Enter registration ID: 2026EC102

Valid Registration ID

Admission Year: 2026

Branch: Electronics

Student Number: 102

Student 3

Enter registration ID: 2025ME103

Valid Registration ID

Admission Year: 2025

Branch: Mechanical

Student Number: 103

Student 4

Enter registration ID: 2026XX104

Invalid Registration ID

====== STUDENT RECORDS ======

2026CS101 2026 CS 101 Computer Science

2026EC102 2026 EC 102 Electronics

2025ME103 2025 ME 103 Mechanical

====== BRANCH CLASSIFICATION ======

Branch: Computer Science

2026CS101

Total Students: 1

Branch: Electronics

2026EC102

Total Students: 1

Branch: Mechanical

2025ME103

Total Students: 1

Branch: Electrical

Total Students: 0

Branch: Information Technology

Total Students: 0

====== ADMISSION YEAR CLASSIFICATION ======

Admission Year: 2026

2026CS101

2026EC102

Admission Year: 2025

2025ME103

====== FINAL SUMMARY ======

Valid Records: 3

Invalid Records: 1

Total Records: 4

7. Flowchart

#chatgpt-mermaid-_r_29v_{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:16px;fill:rgb(237, 237, 237);}@keyframes edge-animation-frame{from{stroke-dashoffset:0;}}@keyframes dash{to{stroke-dashoffset:0;}}#chatgpt-mermaid-_r_29v_ .edge-animation-slow{stroke-dasharray:9,5!important;stroke-dashoffset:900;animation:dash 50s linear infinite;stroke-linecap:round;}#chatgpt-mermaid-_r_29v_ .edge-animation-fast{stroke-dasharray:9,5!important;stroke-dashoffset:900;animation:dash 20s linear infinite;stroke-linecap:round;}#chatgpt-mermaid-_r_29v_ .error-icon{fill:rgb(48, 48, 48);}#chatgpt-mermaid-_r_29v_ .error-text{fill:rgb(237, 237, 237);stroke:rgb(237, 237, 237);}#chatgpt-mermaid-_r_29v_ .edge-thickness-normal{stroke-width:1px;}#chatgpt-mermaid-_r_29v_ .edge-thickness-thick{stroke-width:3.5px;}#chatgpt-mermaid-_r_29v_ .edge-pattern-solid{stroke-dasharray:0;}#chatgpt-mermaid-_r_29v_ .edge-thickness-invisible{stroke-width:0;fill:none;}#chatgpt-mermaid-_r_29v_ .edge-pattern-dashed{stroke-dasharray:3;}#chatgpt-mermaid-_r_29v_ .edge-pattern-dotted{stroke-dasharray:2;}#chatgpt-mermaid-_r_29v_ .marker{fill:rgb(175, 175, 175);stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_29v_ .marker.cross{stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_29v_ svg{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:16px;}#chatgpt-mermaid-_r_29v_ p{margin:0;}#chatgpt-mermaid-_r_29v_ .label{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_29v_ .cluster-label text{fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_29v_ .cluster-label span{color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_29v_ .cluster-label span p{background-color:transparent;}#chatgpt-mermaid-_r_29v_ .label text,#chatgpt-mermaid-_r_29v_ span{fill:rgb(237, 237, 237);color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_29v_ .node rect,#chatgpt-mermaid-_r_29v_ .node circle,#chatgpt-mermaid-_r_29v_ .node ellipse,#chatgpt-mermaid-_r_29v_ .node polygon,#chatgpt-mermaid-_r_29v_ .node path{fill:rgb(9, 23, 44);stroke:rgb(31, 78, 148);stroke-width:1px;}#chatgpt-mermaid-_r_29v_ .rough-node .label text,#chatgpt-mermaid-_r_29v_ .node .label

text,#chatgpt-mermaid-_r_29v_ .image-shape .label,#chatgpt-mermaid-_r_29v_ .icon-shape .label{text-anchor:middle;}#chatgpt-mermaid-_r_29v_ .node .katex path{fill:#000;stroke:#000;stroke-width:1px;}#chatgpt-mermaid-_r_29v_ .rough-node .label,#chatgpt-mermaid-_r_29v_ .node .label,#chatgpt-mermaid-_r_29v_ .image-shape .label,#chatgpt-mermaid-_r_29v_ .icon-shape .label{text-align:center;}#chatgpt-mermaid-_r_29v_ .node.clickable{cursor:pointer;}#chatgpt-mermaid-_r_29v_ .root .anchor path{fill:rgb(175, 175, 175)!important;stroke-width:0;stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_29v_ .arrowheadPath{fill:rgb(175, 175, 175);}#chatgpt-mermaid-_r_29v_ .edgePath .path{stroke:rgb(175, 175, 175);stroke-width:1px;}#chatgpt-mermaid-_r_29v_ .flowchart-link{stroke:rgb(175, 175, 175);fill:none;}#chatgpt-mermaid-_r_29v_ .edgeLabel{background-color:rgb(0, 0, 0);text-align:center;}#chatgpt-mermaid-_r_29v_ .edgeLabel p{background-color:rgb(0, 0, 0);}#chatgpt-mermaid-_r_29v_ .edgeLabel rect{opacity:0.5;background-color:rgb(0, 0, 0);fill:rgb(0, 0, 0);}#chatgpt-mermaid-_r_29v_ .labelBkg{background-color:rgba(0, 0, 0, 0.5);}#chatgpt-mermaid-_r_29v_ .cluster rect{fill:rgb(48, 48, 48);stroke:rgba(255, 255, 255, 0.15);stroke-width:1px;}#chatgpt-mermaid-_r_29v_ .cluster text{fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_29v_ .cluster span{color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_29v_ div.mermaidTooltip{position:absolute;text-align:center;max-width:200px;padding:2px;font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:12px;background:rgb(48, 48, 48);border:1px solid rgba(255, 255, 255, 0.15);border-radius:2px;pointer-events:none;z-index:100;}#chatgpt-mermaid-_r_29v_ .flowchartTitleText{text-anchor:middle;font-size:18px;fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_29v_ rect.text{fill:none;stroke-width:0;}#chatgpt-mermaid-_r_29v_ .icon-shape,#chatgpt-mermaid-_r_29v_ .image-shape{background-color:rgb(0, 0, 0);text-align:center;}#chatgpt-mermaid-_r_29v_ .icon-shape p,#chatgpt-mermaid-_r_29v_ .image-shape p{background-color:rgb(0, 0, 0);padding:2px;}#chatgpt-mermaid-_r_29v_ .icon-shape .label rect,#chatgpt-mermaid-_r_29v_ .image-shape .label rect{opacity:0.5;background-color:rgb(0, 0, 0);fill:rgb(0, 0, 0);}#chatgpt-mermaid-_r_29v_ .label-icon{display:inline-block;height:1em;overflow:visible;vertical-align:-0.125em;}#chatgpt-mermaid-_r_29v_ .node .label-icon path{fill:currentColor;stroke:revert;stroke-width:revert;}#chatgpt-mermaid-_r_29v_ .node .neo-node{stroke:rgb(31, 78, 148);}#chatgpt-mermaid-_r_29v_ [data-look="neo"].node rect,#chatgpt-mermaid-_r_29v_ [data-look="neo"].cluster rect,#chatgpt-mermaid-_r_29v_ [data-look="neo"].node polygon{stroke:url(#chatgpt-mermaid-_r_29v_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_29v_ [data-look="neo"].swimlane.cluster rect{filter:none;}#chatgpt-mermaid-_r_29v_ [data-look="neo"].node path{stroke:url(#chatgpt-mermaid-_r_29v_-gradient);stroke-width:1px;}#chatgpt-mermaid-_r_29v_ [data-look="neo"].node .outer-path{filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_29v_ [data-

look="neo"].node .neo-line path{stroke:rgb(31, 78, 148);filter:none;}#chatgpt-mermaid-_r_29v_ [data-look="neo"].node circle{stroke:url(#chatgpt-mermaid-_r_29v_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_29v_ [data-look="neo"].node circle .state-start{fill:#000000;}#chatgpt-mermaid-_r_29v_ [data-look="neo"].icon-shape .icon{fill:url(#chatgpt-mermaid-_r_29v_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_29v_ [data-look="neo"].icon-shape .icon-neo path{stroke:url(#chatgpt-mermaid-_r_29v_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_29v_ .node text{font-size:14px;font-weight:600;letter-spacing:normal;fill:rgb(153, 206, 255);}#chatgpt-mermaid-_r_29v_ .edgeLabels text{font-size:13px;font-weight:600;letter-spacing:-0.08px;fill:rgb(153, 206, 255);}#chatgpt-mermaid-_r_29v_ .node tspan[font-weight="normal"],#chatgpt-mermaid-_r_29v_ .edgeLabels tspan[font-weight="normal"]{font-weight:600;}#chatgpt-mermaid-_r_29v_ .edgeLabel .label rect{opacity:1;rx:13px;ry:13px;fill:rgb(0, 14, 26);stroke:rgb(26, 62, 95);stroke-width:1px;}#chatgpt-mermaid-_r_29v_ .node rect,#chatgpt-mermaid-_r_29v_ .node circle,#chatgpt-mermaid-_r_29v_ .node ellipse,#chatgpt-mermaid-_r_29v_ .node polygon,#chatgpt-mermaid-_r_29v_ .node path{fill:rgb(0, 40, 77);stroke:rgba(255, 255, 255, 0.1);stroke-width:1px;}#chatgpt-mermaid-_r_29v_ .node rect{rx:16px;ry:16px;}#chatgpt-mermaid-_r_29v_ .mermaid-decision .label-container{fill:rgb(0, 14, 26);stroke:rgb(26, 62, 95);stroke-dasharray:2,2;}#chatgpt-mermaid-_r_29v_ .edgePaths .flowchart-link{stroke:rgb(175, 175, 175);stroke-width:1px;stroke-linecap:round;stroke-linejoin:round;}#chatgpt-mermaid-_r_29v_ .marker{fill:rgb(175, 175, 175);stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_29v_ :root{--mermaid-font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";}StartEnter number of studentsEnter registration IDLength is 9?Mark as invalidYear and number are digits?Valid branch code?Extract year, branch andnumberStore student detailsMore students?Classify by branch and yearDisplay student recordsStopNoYesNoYesNoYesYesNo

8. Concepts Used from Syllabus

Syllabus concept Application

Variables	Registration ID, year
Data types	String and integer
Operators	Comparison operators
Conditional statements	ID validation
Loops	Processing student records
Strings	String slicing and length
Lists	Storing student records
Dictionaries	Branch code mapping
Tuples	Storing extracted details
Tuple unpacking	Extracting student information

9. Conclusion

The Campus Registration Identifier Analyzer validates student registration IDs and extracts important information such as admission year, branch code, and student number. It classifies student records according to their branches and admission years. This program reduces manual work and helps universities organize student registration information efficiently.

Note: The registration format and branch codes used in this program are assumed for educational purposes. They can be modified according to university requirements.